

Marta Marchiori Manerba *(she/her)*

2° Year PhD Student in AI for Society

About

Marta Marchiori Manerba (she/her) is a Graduate Student in Digital Humanities at the University of Pisa and currently a Ph.D. student in AI. During her studies, she explored the relationship between technology and human rights. She works on Fairness and Explainability in Natural Language Processing, focusing on digital discrimination and algorithmic biases. She holds a Bachelor's Degree in Digital Humanities from the University of Pisa, during which she developed a strong interest in hate speech detection towards minorities in online discourse.

Contacts

Born in Mantova, Italy, on 11/3/96 •
Based in Pisa, Italy •
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Research Interests

NLP; XAI; Fairness in ML; Algorithmic bias;
Algorithmic Auditing; ML Evaluation;
Data awareness; Digital Discrimination;
Intersectionality; Hate Speech Detection;
Abusive Language Detection Systems

Education

University of Pisa, Computer Science Department - **National PhD in AI for Society**

► Supervisors: Dr. Riccardo Guidotti; Prof. Salvatore Ruggieri

Doctoral fellowship on *"Science and technology of interpretable machine learning and explanation of AI-assisted decision making"*

NOVEMBER 2021 - ONGOING

University of Pisa - **Master's Degree** in Digital Humanities: Language Technologies

► Supervisor: Dr. Riccardo Guidotti. Mark: 110/110 with honors

Thesis project on Fairness Auditing in Abusive Language Detection Systems carried out in the field of *Explainability and Fairness in ML*

FEBRUARY 2019 - JULY 2021

University of Pisa - **Bachelor's Degree** in Digital Humanities

► Supervisor: Dr. Maria Claudia Buzzi. Mark: 109/110

Thesis project carried out in the field of *Assistive Technology for Education*: development of a website for the research of tools and ICT (Information and Communications Technology) resources to support cognitive training.

OCTOBER 2015 - FEBRUARY 2019

Research and Academic Experience

Supervisor of a Master Thesis in Digital Humanities entitled "Genetic Fairness-Enhancing Data Generation Framework" with Riccardo Guidotti and Martina Cinquini

JANUARY-JULY 2022

Reviewer for LREC Workshop on Perspectivist Approaches to NLP ([NLPerspectives](#))

2022

AIUCD2021 - Assistant moderator during the talk sessions

20-22 JANUARY 2021

Digital Humanities group, Fondazione Bruno Kessler - Traineeship

► Fairness analysis of abusive language detection systems detecting unintended models biases with [CheckList](#): creation of synthetic linguistic data to test a range of social prejudices (e.g. sexism, racism and ableism)

OCTOBER - DECEMBER 2020

University of Pisa - Academic Tutor of the course *Programming and Theoretical Fundamentals*

► Assistance during the laboratory: preparing exercises and assignments, helping students with coding during classes, clarifying basic constructs and concept in Javascript

OCTOBER - DECEMBER 2020

Institute of Computer Science and Telematics, CNR of Pisa - Traineeship in *Assistive Technology for Education*

► Mapping and categorization of ICT resources to support cognitive training through the construction of an inventory of tools available on the internet; Implementation of a digital catalogue to make the collection usable

MAY - SEPTEMBER 2018

Publications

Conference Publications

Marta Marchiori Manerba and Riccardo Guidotti. "[Investigating Debiasing Effects on Classification and Explainability](#)". AAAI/ACM Conference on Artificial Intelligence, Ethics, and Society ([AIES](#)), 2022

In-person presentation

Marta Marchiori Manerba and Riccardo Guidotti. "[FairShades: Fairness Auditing via Explainability in Abusive Language Detection Systems](#)". IEEE Conference on Cognitive Machine Intelligence ([CogMI](#)), 2021.

Online presentation, Best Paper Award

Workshop Publications

Marta Marchiori Manerba and Virginia Morini. "Exposing Racial Dialect Bias in Abusive Language Detection: Can Explainability Play a Role?". ECML PKDD International Workshop on eXplainable Knowledge Discovery in Data Mining ([XKDD](#)), 2022

In-person presentation

Marta Marchiori Manerba, Riccardo Guidotti, Lucia Passaro, Salvatore Ruggieri. "[Bias Discovery Within Human Raters: A Case Study of the Jigsaw Dataset](#)". LREC Workshop on Perspectivist Approaches to NLP ([NLPerspectives](#)), 2022

In-person presentation

Arianna Muti, Marta Marchiori Manerba, Katerina Korre, and Alberto Barrón-Cedeño. "[LeaningTower@LT-EDI-ACL2022: When Hope and Hate Collide](#)". ACL Workshop on Language Technology for Equality, Diversity and Inclusion ([LT-EDI](#)), 2022

Marta Marchiori Manerba and Sara Tonelli. "[Fine-Grained Fairness Analysis of Abusive Language Detection Systems with CheckList](#)". Workshop on Online Abuse and Harms ([WOAH!](#)), 2021

Online presentation

Federico Mazzoni, Marta Marchiori Manerba, Martina Cinquini, Riccardo Guidotti, and Salvatore Ruggieri. "FairGen: A Genetic Fairness-Enhancing Data Generation Framework". Intelligent Data Analysis (IDA), 2023

Research Dissemination

"[Picture a Scientist](#)": Organisation of a panel discussion financed by funds for PhD students' scientific initiatives. The event aims to analyse the reasons, factors and motives behind the lower percentage of women choosing to pursue education and careers in STEM fields

2022-2023

High school civic education lesson at IIS Cobianchi in Verbania (online): "Ethics in AI Systems: Biases in ML Models"

MAY 2022

Creation of a video for university choice orientation on Digital Humanities and the impacts of AI in the context of the initiative "[A Choice for Life](#)"

MARCH 2022

Intervention at a university orientation event for high schools (held at the CS Department in Pisa): "Ethics in AI Systems and the Timnit Gebru case"

MARCH 2022

Digital Advocacy & Volunteering

[Privacy Network](#) - Team member

► Non-profit association that promotes a culture of privacy and responsible use of technology and that advocates for digital rights

NOVEMBER 2022 - IN PROGRESS

[KRINO](#) - Vice president and board member (2022)

► Cultural association that organizes panels, workshops and events within the field of Digital Humanities, with the aim of integrating and contaminating sciences and arts

- Active participation in projects of critical divulgation of topics such as: computational thinking, datafication, digital culture, demystification of AI systems, digital art, live coding
- Collaborative and collective writing and editing of posts, articles, newsletters
- Internal training sessions

NOVEMBER 2019 - IN PROGRESS